Sustainable Recipe Transformation through Multi-Objective Ingredient Optimization

1. Problem Statement

Food choices have a significant environmental impact, and ingredients within a recipe can contribute very differently to the overall carbon footprint. At the same time, simply replacing a high-impact ingredient with a low-impact ingredient may alter the flavor profile, nutritional composition, or culinary characteristics of the original recipe.

CoSyLab has already developed computational resources for recipes, ingredients, flavor molecules, nutrition and sustainability, along with previous work on recipe generation and recipe transformation. This project proposes to build on these existing resources and investigate a more focused problem:

Can an existing recipe be transformed through targeted ingredient substitutions that reduce environmental impact while preserving its flavor characteristics and nutritional requirements?

For example, given a recipe containing beef, the system could identify suitable alternative ingredients and determine an optimized substitution while considering:

- Reduction in carbon footprint
- Molecular flavor similarity
- Preservation of nutritional properties
- Ingredient compatibility and recipe structure

The goal is therefore not simply to generate a new recipe, but to **optimize a transformation of an existing recipe**.

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2. Relation to Existing CoSyLab Work

The project will build upon existing CoSyLab resources and research rather than reproducing them.

Relevant existing work includes:

- **RecipeDB2** for structured recipe, ingredient and nutritional information
- **FlavorDB2** for ingredient–flavor molecule relationships
- **Carbon Footprint** resources for estimating the environmental impact of recipes
- Previous work on **recipe generation**
- Previous work on **Cuisine Fusion**, which explored transforming recipes through ingredient substitutions

The 2023 Cuisine Fusion BTP specifically investigated transforming a recipe from a source cuisine to a target cuisine by replacing ingredients and subsequently classifying the transformed recipe.

This project differs in its primary objective. Instead of transforming a recipe according to cuisine, the proposed system will investigate **environmentally motivated ingredient substitution as a multi-objective optimization problem**, while explicitly measuring the trade-off between environmental impact, flavor preservation and nutrition.

The project will therefore treat existing CoSyLab systems as building blocks and concentrate on the optimization and evaluation layer.

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3. Research Objective

The primary objective is to develop and evaluate a computational framework that can identify ingredient substitutions which provide a favorable trade-off between:

Environmental Impact ↓

Flavor Divergence ↓

Nutritional Deviation ↓

while maintaining the structural and culinary characteristics of the original recipe.

The project will investigate the following research question:

> **How can ingredient substitutions be optimized to reduce the environmental impact of an existing recipe while maintaining acceptable flavor similarity and nutritional properties?**

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4. Proposed Methodology

Phase 1: Recipe and Ingredient Representation

We will use existing structured computational gastronomy resources to represent recipes and ingredients.

For an input recipe, the system will obtain:

- Ingredients and quantities
- Nutritional information
- Carbon-footprint information
- Flavor molecules associated with ingredients
- Ingredient co-occurrence information

The existing CoSyLab ecosystem already provides structured resources for recipes, flavors, nutrition and sustainability, making it possible to focus the project on the transformation and optimization problem rather than rebuilding these databases.

Phase 2: Candidate Ingredient Generation

For each ingredient that is a potential substitution candidate, the system will construct a set of possible replacement ingredients.

Candidate generation will consider:

1. Flavor similarity

Ingredients sharing similar molecular flavor profiles will be considered stronger candidates.

Flavor similarity can be calculated using representations derived from FlavorDB2 and an appropriate similarity metric.

2. Ingredient co-occurrence

The recipe database can be used to identify ingredients that commonly occur in similar culinary contexts.

This provides a measure of whether a candidate ingredient is compatible with the surrounding ingredients.

3. Environmental impact

Candidate ingredients with lower carbon footprints will be preferred when they can satisfy the other requirements.

4. Nutritional characteristics

The nutritional composition of the transformed recipe will be compared with the original recipe to avoid unacceptable nutritional changes.

This produces a candidate substitution graph:

Original Ingredient → Candidate Ingredients

with each candidate associated with:

- Carbon impact
- Flavor similarity
- Nutritional difference
- Compatibility/co-occurrence score

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5. Multi-Objective Optimization

The main research contribution will be the formulation of ingredient substitution as a multi-objective optimization problem.

For a transformed recipe (R'), we define three primary objectives:

Objective 1 — Environmental Impact

Minimize the carbon footprint:

Minimize C(R′)

where C(R′) represents the total carbon footprint of the transformed recipe.

Objective 2 — Flavor Divergence

Minimize the difference between the original and transformed flavor profiles:

Minimize D_f(R, R′)

where D_f(R, R′) represents the difference between the molecular flavor profiles of the original recipe R and the transformed recipe R′.

Objective 3 — Nutritional Deviation

Minimize the deviation of important nutritional properties from the original recipe or from specified nutritional targets:

Minimize D_n(R, R′)

where D_n(R, R′) represents the nutritional difference between the original and transformed recipes.

Overall Optimization

Minimize [C(R′), D_f(R, R′), D_n(R, R′)]

subject to recipe-level constraints such as:

- Acceptable flavor similarity
- Required protein or calorie range
- Ingredient compatibility
- Valid ingredient quantities
- Valid recipe composition

Rather than assuming that one fixed weighting is universally correct, the project can investigate **Pareto-optimal solutions** and study the trade-off between sustainability, flavor and nutrition.

This would allow the system to provide alternatives such as:

Maximum carbon reduction

Best flavor preservation

Best balanced solution

rather than producing only one arbitrary answer.

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6. Explainability

The system should provide an explanation for every major substitution.

For example:

Beef → Mushroom + Lentil blend

because:

- Carbon footprint decreased by X%
- Flavor similarity = X
- Protein changed by X%
- The replacement ingredients have strong compatibility with the remaining recipe ingredients

This makes the optimization process interpretable rather than treating it as a black-box recommendation system.

The explanation layer will also allow us to study **why a particular ingredient was selected** and which objective caused one candidate to outperform another.

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7. Evaluation

The project will evaluate the transformation engine using existing recipes as test cases.

For each recipe, we will compare:

Original Recipe

against

Optimized Recipe

using quantitative measures such as:

- Percentage reduction in carbon footprint
- Flavor similarity score
- Nutritional deviation
- Number of substitutions
- Ingredient compatibility
- Optimization time

We will also compare different optimization strategies to determine how the resulting recipes change when different objectives are prioritized.

For example:

Carbon-focused optimization

vs.

Flavor-focused optimization

vs.

Balanced multi-objective optimization

This will allow us to study the actual trade-offs between sustainability and preservation of culinary characteristics.

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8. Research Timeline

Semester 1 — Data, Representation and Baseline

Work

- Study and use existing CoSyLab datasets and APIs
- Understand RecipeDB2, FlavorDB2 and carbon-footprint data
- Construct recipe and ingredient representations
- Develop ingredient candidate-generation methods
- Build flavor similarity and ingredient compatibility measures
- Implement a baseline substitution system
- Establish evaluation metrics

Deliverable

A working **Ingredient Substitution and Sustainability Scoring System** that can take an existing recipe, identify candidate substitutions and quantitatively compare them with the original recipe.

The first semester will establish the complete data and evaluation pipeline required for optimization.

Semester 2 — Optimization, Evaluation and Web Interface

Work

- Formulate ingredient substitution as a multi-objective optimization problem
- Implement and compare optimization strategies
- Generate Pareto-optimal recipe transformations
- Evaluate carbon, flavor and nutritional trade-offs
- Develop an explainability layer
- Build a web interface for demonstration
- Conduct systematic experiments on a representative recipe set
- Document findings and prepare the final thesis

Final Deliverable

A working **Sustainable Recipe Transformation Engine** that:

1. Takes an existing recipe as input
2. Identifies possible ingredient substitutions
3. Optimizes substitutions according to carbon, flavor and nutrition
4. Produces one or more optimized recipe alternatives
5. Quantitatively reports the trade-offs
6. Explains why each substitution was selected
7. Provides an interactive web interface for demonstration

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9. Expected Research Contribution

The project is not intended to recreate existing CoSyLab databases or existing recipe-generation systems.

Instead, the proposed contribution is a focused investigation of:

Multi-objective ingredient substitution for sustainable recipe transformation

The project will study how existing computational representations of recipes, flavor and sustainability can be combined with optimization techniques to answer a specific question:

> **What is the lowest-impact transformation of an existing recipe that remains sufficiently similar in flavor and nutrition?**

The research will also investigate the trade-offs between these objectives rather than optimizing carbon footprint independently.

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10. Why This is a Suitable Extension of CoSyLab Research

The project naturally builds on the computational gastronomy ecosystem developed by CoSyLab.

Existing lab resources already provide the necessary computational foundations:

RecipeDB2 → Recipes and ingredients

FlavorDB2 → Molecular flavor information

Carbon Footprint → Environmental impact

The lab has also previously explored recipe generation and recipe transformation.

The 2025 doctoral thesis under Prof. Ganesh Bagler further demonstrates the broader research direction connecting recipes, flavors, nutrition and sustainability.

Therefore, the proposed work is intended as a **focused next step**, concentrating on the optimization of ingredient substitutions and quantitative evaluation of the resulting sustainability–flavor–nutrition trade-offs.

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11. Possible Future Extensions

If the core optimization framework is successful, the system can later be extended to:

- Water-footprint optimization
- Cost-aware ingredient substitution
- Regional ingredient availability
- Personalized nutritional constraints
- Dietary restrictions
- Cuisine-specific transformation
- LLM-based generation of cooking instructions for transformed recipes
- Human evaluation of taste and culinary acceptability

These extensions will remain outside the initial core implementation unless required by the research findings.

12. Summary

The proposed project will develop a computational framework for transforming existing recipes through **multi-objective ingredient optimization**.

Rather than generating arbitrary new recipes, the system will start from an existing recipe and search for substitutions that achieve a favorable balance between:

Lower environmental impact + Preserved flavor + Preserved nutrition

The project will leverage the existing computational gastronomy infrastructure of CoSyLab and focus its research contribution on **candidate substitution, optimization, trade-off analysis and explainable recipe transformation**.